

Generative Economic Modeling*

Incomplete and preliminary

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Abstract

We introduce a novel approach to solving quantitative economic models: generative economic modeling. Our method builds jointly on neural networks and conventional solution methods. Specifically, we employ neural networks to learn from simplified or partial versions of an economic model and subsequently generate the full model's dynamic behavior. By relying on these less complex satellite models, we circumvent the curse of dimensionality, allowing for the use of well-established conventional solution methods. We show that our approach solves economic models featuring distinct nonlinear dynamics as well as heterogeneous agents.

Keywords— Machine Learning, Neural networks, Nonlinearities, Heterogeneous Agents

JEL codes— C11, C45, D31, E32, E52

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1 Introduction

The advances in artificial intelligence provide substantial opportunities for quantitative economics by shifting the production-possibility frontier of modeling. Deep learning has emerged as a powerful tool for solving dynamic economic models that were previously considered intractable. The curse of dimensionality - first articulated by [Bellman \(1957\)](#) - typically limits the complexity of economic models to only a few state variables. However, deep learning can help to tame this problem, as discussed in [Fernández-Villaverde, Nuño and Perla \(2024\)](#). Unfortunately, successfully employing deep learning in practice often necessitates meticulous and detailed adjustments tailored to the specific model at hand, which makes it challenging to employ. In contrast, established conventional solution methods, though constrained by the curse of dimensionality, are already specifically designed and optimized for particular types of economic models and feature well-understood emergence properties. To combine the strengths of both artificial intelligence and conventional solution methods, we propose a novel approach: generative economic modeling.

Our approach uses neural networks to approximate the full economic model. However, instead of using the complete model as input for the training process, the neural network learns from a collection of simplified models, which we refer to as “satellite models”. Each satellite model includes only a subset of features and shocks, featuring only partially the dynamics of the full model.¹ By relying on these satellite models, we can solve these simpler models with conventional methods without running in the curse of dimensionality. We can then use the solved model to simulate it. The generated simulated data is used to train the neural network to learn the dynamics of the model. While each satellite model captures only some features of the underlying economic model, we ensure an overlap of features in the satellite models. The neural network is then trained on this data to learn to approximate the solution of a complete model that includes all features and shocks.

Our method belongs to the class of generative artificial intelligence because we employ the neural network to generate results for the complete model that includes all features and shocks, something we have not used for the training process. Generative artificial intelligence has achieved significant success, especially in the context of large language models, which are very large deep learning models. While this success also holds promise for our approach, the generative performance of neural networks in our context of economic modeling is less clear.

In this paper, we establish that we can use our method for quantitative economic mod-

¹ Most researchers are used to working with a satellite version of the model, as computational requirements often force them to limit the size of their model. Our approach formalizes how to use a set of satellite models and apply deep learning techniques to work with a more complex version of the model.

els that feature nonlinear dynamics or heterogeneous agents. We first focus on practical demonstrations of our method to solve a variety of models with increasing complexity. We benchmark our new methodology against analytical solutions or widely used alternative global solution methods. Our reference model is a real business cycle model, where we work with three versions: i) a simplified version that can be solved analytically, ii) a medium-sized version with state-dependent investment costs that result in distinct nonlinearities, and iii) a heterogeneous agents version with partially uninsurable income risk in line with [Krusell and Smith \(1998\)](#). We construct for all these models overlapping satellite models and then use simulated data to train our neural network. Thus, we can evaluate how well our approach is designed to capture nonlinearities and how applicable it is to the class of heterogeneous agent versions.

Our methodology provides a very good approximation in all three versions of the model. When evaluated against the true model, our approach consistently yields low prediction errors below one percent, highlighting its robustness and accuracy. Notably, the magnitude of these errors is comparable to that of the neural network trained on data generated by the full model. This holds for the linear aggregate dynamics in the analytical model, the heterogeneous agent version as well as the nonlinear dynamics in the medium-sized model with piecewise nonlinear capital adjustment costs, highlighting its flexibility. Importantly, the latter result underscores the potential of our method for addressing models with higher-order nonlinearities, which typically require computationally intensive global solution techniques. To summarize, our generative economic modeling approach effectively approximates the dynamics of the reference model with sufficient precision. This finding underscores the potential of our approach in efficiently learning and replicating complex model dynamics from simplified model versions.

When solving the heterogeneous agent model, we leverage an additional advantage of our methodology that further reduces the numerical burden. Heterogeneous agent models with multiple aggregate shocks face a threefold curse of dimensionality. First, expanding the state space significantly increases the computational complexity of solving the household problem. Second, incorporating additional shocks complicates the accurate computation of expected values. Third, introducing more shocks necessitates a more involved calculation of the perceived laws of motion to forecast the future evolution of aggregate variables. Our methodology addresses these challenges by effectively reducing the state space by only solving satellite models, enabling researchers to solve more intricate models with greater efficiency. By accelerating the solution process for complicated models and enabling the analysis of yet unsolvable economic environments, our approach provides a powerful tool for advancing research on nonlinear and high-dimensional macroeconomic

models, especially in the heterogeneous agent literature.

Another important feature of our method is that we can cross-check its fit using metrics such as the Euler equation errors. Such metrics evaluate how well the neural network can approximate the results. Therefore, the fit of the method can be evaluated also when we move away from the laboratory setting provided above. These metrics also help set up the satellite models and specify the neural network. Instead of providing a clear handbook or proof, our approach follows the data science literature. We evaluate which combination of satellite models and neural networks provides the best approximation and use this specification for our problem at hand.

As initial motivation for our approach, we highlighted the potential fragility of deep learning when used directly to solve the economic model. However, our approach does not feature this problem because of a difference in the type of training data. The data to train in our approach is not affected by the neural network training because it is precalculated and comes from the conventional solution method. When deep learning is directly used to solve economic equations, the inputs used depend on the model solution from the neural network. Therefore, the inputs are endogenous (instead of exogenous) because they depend on the neural network. This feedback complicates finding the solution substantially, an issue that is absent in our generative economic modeling approach.

Literature Review Our paper belongs to the fast-growing literature that uses deep learning to solve dynamic economic models. The areas of application have been HANK models (Fernández-Villaverde et al., 2024; Kase, Melosi and Rottner, 2022), heterogeneous agents (Azinovic, Gaegauf and Scheidegger, 2022; Fernández-Villaverde, Hurtado and Nuno, 2023; Gorodnichenko et al., 2021; Han, Yang and E, 2021; Kahou et al., 2021; Maliar and Maliar, 2022; Maliar, Maliar and Winant, 2021), overlapping generations and life-cycle (Azinovic and Zemlicka, 2024; Druedahl and Røpke, 2025; Pascal, 2024), finance (Chen, Didisheim and Scheidegger, 2023; Duarte, Duarte and Silva, 2024; Duarte and Fonseca, 2024), labor markets and search (Adenbaum, Babalievsky and Jungerman, 2024; Jungerman, 2024; Payne, Rebei and Yang, 2024), monetary policy (Chen et al., 2021; Nuño, Renner and Scheidegger, 2024), climate change (Fernández-Villaverde, Gillingham and Scheidegger, 2024; Friedl et al., 2023), and behavioral macroeconomics (Ashwin, Beaudry and Ellison, 2025; Kahou et al., 2024). However, our approach to using neural networks deviates strongly from these papers, as we are not interested in directly solving the economic equations. More closely related to our work is the usage of neural networks as surrogate models as in Kase, Melosi and Rottner (2022) and Chen, Didisheim and Schei-

degger (2023).² Yet, these papers use the complete underlying model for the training, thereby excluding the generative part. Finally, we also differ by following a hybrid approach that exploits the advantages of conventional solution methods and deep learning.

Our work also builds on the broader literature on conventional solution methods that do not rely on deep learning. Given the vast array of contributions across different fields of computational economics, providing a comprehensive review would be infeasible. Instead, we refer to the influential books on numerical methods by Judd (1998), Miranda and Fackler (2004) and Heer and Maussner (2024), which provide a great overview of the methods available. Our approach builds upon these traditional methods while leveraging deep learning to enhance their capacity to handle higher levels of complexity. The combination of the methods makes it possible to tackle problems that were previously computationally intractable.

2 Generative Economic Modeling

This section outlines our generative economic modeling approach, which is designed to solve a large class of dynamic general equilibrium models.³

2.1 Underlying Dynamic General Equilibrium Model

The dynamics of a dynamic general equilibrium model can be expressed as a transition equation:

$$\mathbb{S}_t = f(\mathbb{S}_{t-1}, \nu_t | \Theta) \quad (1)$$

where the state vector $\mathbb{S}_t \in \mathbb{R}^m$ describes the economy in the economy in period t . Note that such a representation can contain heterogeneous agents or behavioral models. The economy is also subject to exogenous shocks that follow a Markov process, which is captured by the vector $\nu_t \in \mathbb{R}^n$. There is also a vector of structural parameters $\Theta \in \mathbb{R}^d$, which affects the dynamics of the model. The function $f(\cdot)$ determines the mapping from the previous period state variables $\mathbb{S}_{t-1}$ and current period shocks ν_t to the current period state variables $\mathbb{S}_t$ conditional on the structural parameters. To solve the transition equation, it is needed to find the policy function (decision function) which maps the model state variables $\mathbb{S}_t$ to a set of choices ψ_t .

² A surrogate model is an approximate model that mimics the behavior of a more complex model.

³ We focus on dynamic Markov economic models, where agents solve a Markov decision problem, as e.g. in Maliar, Maliar and Winant, 2021 and Fernández-Villaverde, Nuño and Perla, 2024.

This mapping is usually unknown and needs to be solved with numerical methods. Luckily, there already exists a large set of solution methods - more general solution approaches, like value function iteration or policy function iteration, and very tailored solution methods, such as the Krusell-Smith approach for heterogenous agent economies with aggregate risk. In general, the idea is to find an equilibrium function that maps the state variables to a set of control variables, $\psi_t = \psi(\mathbb{S}_t|\Theta)$. These policy functions satisfy a set of equations derived from the model.

$$\mathbb{F}(\psi_t(\cdot)) = 0 \quad (2)$$

Once equipped with these policy functions, we can solve for the transition equation.

The advantage of our approach is that we do not attempt to innovate on this dimension, and we leave the conventional solution steps unchanged as they have been tailored for years or even decades for specific problems. However, such conventional global solution methods face the curse of dimensionality. The exponential growth of the state space as the number of states increases limits the complexity of the economic model that can be considered. For instance, solution methods that use full grid-based approaches require N^{m+n} points per dimension, which results in exponential computing costs, as shown by the order of the integration error $O(N^{m+n})$.⁴

2.2 Satellite Models of the Complete Model

The curse of dimensionality often forces the modeler to reduce the complexity of the studied model by limiting the number of state variables and shocks. In other words, a simplified satellite model is derived from the underlying complete model. In practice, using a satellite model instead of the most comprehensive model available is mostly the norm when working with global solution methods.⁵ In that regard, economists are well-trained to use satellite models, and it is likely the common approach.

We denote the variables in the satellite version with a $\sim$ and rewrite the transition equation of the satellite model as:

$$\tilde{\mathbb{S}}_t = \tilde{f}(\tilde{\mathbb{S}}_{t-1}, \tilde{\nu}_t|\tilde{\Theta}) \quad (3)$$

where the dimension of the states $\tilde{\mathbb{S}}_t \in \mathbb{R}^{\tilde{m}}$ and/or shocks $\tilde{\nu}_t \in \mathbb{R}^{\tilde{n}}$ is smaller than in the full

⁴ Note that refinements to the solution method can lower the computational costs, e.g. adaptive sparse grids (see [Brumm and Scheidegger \(2017\)](#)).

⁵ Models that are solved with perturbation methods are usually much larger than models solved with global solution methods.

model, that is $(\tilde{m} < m) \vee (\tilde{n} < n)$. Note that the set of structural parameters $\tilde{\Theta} \in \tilde{\mathbb{R}}^d$ is then also potentially smaller, that is $\tilde{d} \leq d$.

However, we can now specify not only one satellite model, but instead several satellite models that capture different elements of the underlying model, that is $\tilde{S}_t^a, \tilde{S}_t^b, \tilde{S}_t^c, \dots$, where the superscript indicates the satellite model.

$$\tilde{S}_t^i = \tilde{f}^i \left(\tilde{S}_{t-1}^i, \tilde{\nu}_t^i | \tilde{\Theta}^i \right), \text{ for } i = a, b, c, \dots \quad (4)$$

All these satellite models together form a set $\bar{f}_t$:

$$\bar{f} \left(\tilde{S}_{t-1}, \tilde{\nu}_t | \tilde{\Theta} \right) = \left\{ \tilde{f}^a \left(\tilde{S}_{t-1}^a, \tilde{\nu}_t^a | \tilde{\Theta}^a \right), \tilde{f}^b \left(\tilde{S}_{t-1}^b, \tilde{\nu}_t^b | \tilde{\Theta}^b \right), \tilde{f}^c \left(\tilde{S}_{t-1}^c, \tilde{\nu}_t^c | \tilde{\Theta}^c \right), \dots \right\} \quad (5)$$

Importantly, the numerical costs to extend the set increase linearly instead of exponentially (conditional on keeping the same number of states and shocks for each satellite model). The satellite models can be specified to be complete and overlapping. We define completeness as each state S_t and shock ν_t is at least covered in one satellite model. Therefore, at least one satellite model captures one specific part of the underlying model. This requirement ensures that the set of satellite models and the true model have the same states, shocks, and parameters. We define overlapping as each state S_t and shock ν_t should be at least covered in two different satellite models, so that the different subsets overlap.

2.3 Deep Learning Approach to Reconstruct the Full Model

The set of satellite models is a subset of the true underlying model, that is

$$\bar{f} (S_{t-1}, \nu_t | \Theta) \subseteq f (S_{t-1}, \nu_t | \Theta) \quad (6)$$

The idea of this paper is to now to evaluate whether a sufficient rich specified set of satellite models, to approximate the true underlying model, that is

$$\bar{f} (S_{t-1}, \nu_t | \Theta) \approx f (S_{t-1}, \nu_t | \Theta) \quad (7)$$

Even though the set of satellite models is complete and overlapping, the satellite models are only partial representations of the full models.

For this reason, we want to use deep learning to learn the underlying dynamics of the full model, leveraging on its generative capacity. We approximate the transition equation

using neural network, that is

$$f(\mathbb{S}_{t-1}, \nu_t | \Theta) \approx f_{DNN}(\mathbb{S}_{t-1}, \nu_t | \Theta) \quad (8)$$

where we the subscript DNN indicates that this is a deep neural network.

We would then train the neural network using simulated data. Assuming that we would know the transition equation for the underlying full model, we would draw a sequence of vectors of shocks $\{\nu_t\}_{t=1}^N$. Given these shocks and an initial condition, we update the states $\mathbb{S}_t$ and controls ψ_t period-by-period. This would result in the following dataset:

$$\mathbb{D} = \{\mathbb{S}_t, \psi_t, \nu_t\}_{t=1}^N \quad (9)$$

Using the simulated dataset, we can train a neural network to learn the mapping from previous period state variables $\mathbb{S}_t$ and shocks ν_t to current period state variables $\mathbb{S}_t$, in other words the function $f(\cdot)$. In particular, we train the neural network to minimize the error between its predicted values and simulated values, summarized loss $\mathcal{L}$. The neural network minimizes its weights using the provided data

$$f_{DNN} = \arg \min_W \mathbb{L}(W | \mathbb{D}) \quad (10)$$

However, due to the curse of dimensionality, it is too costly to solve the model so that we resort to our satellite model approach. For this reason, we simulate each satellite model separately. We then merge all simulations, creating a long data series in which only a specific subset of states and socks is active in different periods. The dataset that holds the simulation for all satellite models is:

$$\bar{\mathbb{D}} = \left\{ \left\{ \tilde{S}_t^a, \psi S_t^a, \tilde{\nu}_t^a \right\}_{t=1}^N, \left\{ \tilde{S}_t^b, \psi S_t^b, \tilde{\nu}_t^b \right\}_{t=1}^N, \left\{ \tilde{S}_t^c, \psi S_t^c, \tilde{\nu}_t^c \right\}_{t=1}^N, \dots \right\} \quad (11)$$

Note that because our set of satellite models is complete, this training data covers all states and shocks, that is $\mathbb{S}_t$ and ν_t . We also divide the collected dataset into a training and validation sample.⁶

We are now training the neural network using the datasets from the satellite models by minimizing the mean squared errors between the predicted values from the neural network

⁶ Note that simulated data for each satellite model is distributed between the training and validation sample to ensure that both samples contain the same share of each satellite model.

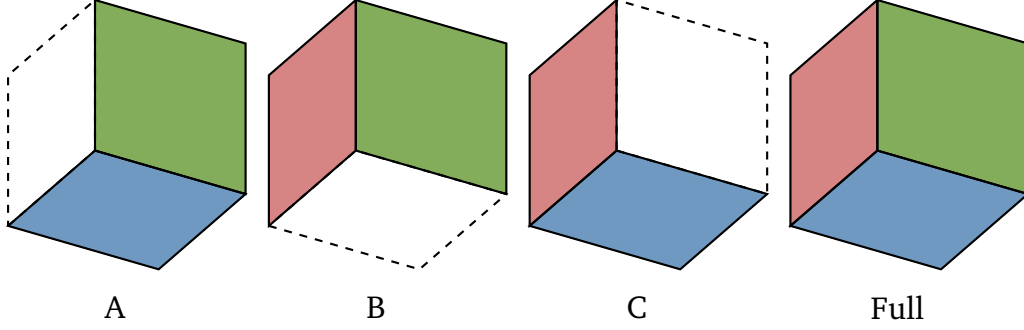


Figure 1 Illustration of Satellite Models and Generic Economic Modeling

and the observed values from the satellite models:

$$\bar{f}_{DNN} = \arg \min_{\bar{W}} \mathbb{L}(\bar{W} | \bar{\mathbb{D}}) \quad (12)$$

By training the network on data from multiple satellite models, it learns the distinct transmission mechanisms of individual shocks, effectively generating an economic model that integrates the key features of the underlying data. This approach allows us to approximate the full model's dynamics by leveraging insights from its components.

For our proof of concepts later, we employ a fully connected deep neural network consisting of three hidden layers, each containing 128 nodes. We train the network using a batch size of 64 for approximately half a million iterations, with an initial learning rate of 10^{-4} , which decreases during training following a cosine annealing schedule with a terminal learning rate of 10^{-7} .

Illustration of our Approach To illustrate our approach with a simple example, imagine an economic model with a total set of shocks $\mathcal{E}$. However, the model is too complex to be solved in its full specification. For this reason, we solve the satellite models, each of which includes only a subset $\tilde{\mathcal{E}}$. The number of required satellite models is determined by the binomial coefficient:

$$\binom{n_{\mathcal{E}}}{n_{\tilde{\mathcal{E}}}} = \frac{n_{\mathcal{E}}!}{n_{\tilde{\mathcal{E}}}!(n_{\mathcal{E}} - n_{\tilde{\mathcal{E}}})!} \quad (13)$$

Figure 1 provides a graphical illustration for a model with three shocks. However, we solve only satellite models that incorporate two shocks, i.e., $n_{\tilde{\mathcal{E}}} = 2$. As a result, each satellite model features a distinct subset of two shocks. For instance, satellite model A includes the blue and green shocks, while others incorporate different combinations. In total, we have three satellite models (A, B, and C), capturing all possible shock combinations. The

choice of solution algorithm is left to the researcher, as the approach is compatible with any method that can solve the model dynamics.⁷

External Validation One issue is that we cannot provide proof that a sufficiently rich set of satellite models approximates the true underlying model. However, a leap of faith is nevertheless not needed. Instead, we can directly employ standard methods to check the approximation error. In particular, we can use selected equilibrium conditions and calculate the associated Euler equation errors.

The equilibrium conditions can be taken directly from the conventional solution step for the full model, which we never solve. However, we now use our mapping from the generative neural network to assess the fit. Note that we can evaluate the expectations using different methods such as Monte Carlo or Gaussian rules, depending on the problem.

Summary The procedure for generative economic modeling unfolds in three steps. We first solve a set of satellite models using conventional solution methods. We then use the set of satellite models to simulate data. Finally, we train a neural network with deep learning techniques using the simulated data from the satellite models.

3 Model Environment

This section illustrates the model environment, which we use as a benchmark to evaluate our method. We choose three different versions of the real business cycle (RBC) model with distortionary taxes to showcase our method: i) a simplified version that can be solved analytically, ii) a medium-sized version with state-dependent investment costs that result in distinct nonlinearities, and iii) a heterogeneous agents version with partially uninsurable income risk in line with [Krusell and Smith \(1998\)](#).

3.1 Real Business Cycle Model with Government

We use an extended stochastic RBC model composed of a firm sector, a household sector, and a government sector to test the predictive power of our method. The firm sector comprises (i) final goods producers who bundle the intermediate goods, (ii) intermediate goods producers who rent out labor services and capital from perfectly competitive markets but

⁷ While any solution method can be used, the methodology yields optimal performance when applied to solution techniques that minimize approximation errors. The surrogate model's accuracy depends on how well the training data represents the true data generation process. In our applications later on, we employ global solution techniques to solve the model as accurately as possible.

face monopolistic competition in the goods market as they produce differentiated goods, and (iii) producers of capital goods who turn final goods into capital subject to adjustment costs. The adjustment costs are state-dependent, which introduces a non-linearity in the model. Prices remain flexible so monopolistic competition only has redistributive effects.

Households earn income from supplying labor n_{it} and capital k_{it} , and earn profits Π_{it} from owning the firm sector. Households spend their income for consumption c_{it} and capital investment k_{it+1} . Households can be heterogeneous in capital holdings k_{it} , in their idiosyncratic income component h_{it} , and in the allocation of profits Π_{it} .

Finally, the government levies distortionary labor- and capital-income taxes with tax rates τ^L and τ^K , besides a value-added tax on consumption τ^C . Raising taxes is purely distortionary since the government returns the tax revenues to the household via lump-sum transfers T_t .

Production sector: The production sector features final, intermediate, and capital goods producers. Final goods producers bundle varieties j of differentiated intermediate goods according to the Dixit-Stiglitz aggregator

$$Y_t = \left(\int y_{jt}^{\frac{1}{\mu_t}} dj \right)^{\mu_t}, \quad (14)$$

with elasticity of substitution $\frac{\mu_t-1}{\mu_t}$. We allow the markup μ_t to evolve stochastically as an AR(1) in its log

$$\ln \mu_t = (1 - \rho_\mu) \left(\ln \mu - \frac{\sigma_\mu^2}{2} \right) + \rho_\mu \ln \mu_{t-1} + \epsilon_t^\mu \quad \text{with} \quad \epsilon_t^\mu \sim N(0, \sigma_\mu^2). \quad (15)$$

The shock ϵ_t^μ is normally distributed with mean zero and variance σ_μ^2 . Firms can adjust prices in each period, hence markup shocks only redistribute between profits and the factor incomes.⁸

Final goods producers purchase a variety of goods from a continuous range of intermediate producers indexed by j . Production of intermediate goods occurs according to constant returns to scale Cobb-Douglas production technology which combines labor N_{jt} and capital services $u_{jt}K_{jt}$ taking into account capital utilization u_{jt} according to

$$Y_{jt} = A_t (u_{jt}K_{jt})^\alpha (Z_t N_{jt})^{1-\alpha}, \quad (16)$$

⁸ Each differentiated good is offered at price p_{jt} , the aggregate price level is $P_t = \left(\int p_{jt}^{\frac{1}{1-\mu_t}} dj \right)^{1-\mu_t}$ and demand for each of the varieties is $y_{jt} = \left(\frac{p_{jt}}{P_t} \right)^{\frac{1-\mu_t}{\mu_t}} Y_t$. In a symmetric equilibrium, this boils down to $p_{jt} = P_t \forall j$ and $y_{jt} = Y_t \forall j$ and we do not need to keep track of prices hereafter.

where α denotes the capital share in the Cobb-Douglas production function, A_t denotes aggregate productivity and Z_t denotes labor-augmenting technology. Firms can choose the intensity with which they use their capital stock K_{jt} by adjusting the capacity utilization u_{jt} . An intensity higher than normal results in increased depreciation of capital according to $\delta(u_{jt}) = \delta_{0t} + \delta_1(u_{jt} - 1) + \delta_2/2(u_{jt} - 1)^2$, which is an increasing and convex function of utilization if $\delta_1, \delta_2 > 0$.

The producer minimizes costs, $w_t N_{jt} - [r_t + q_t \delta(u_{jt})] K_{jt}$, where r_t and q_t are the rental rate and the (producer) price of capital goods and w_t is the real wage. Factor markets are perfectly competitive and all intermediate goods producers are symmetric. Therefore, we drop all indices j and only refer to the aggregate variables. We can characterize the first-order conditions for labor and effective capital as

$$r_t + q_t \delta(u_t) = \frac{\alpha}{\mu_t} A_t u_t \left(\frac{u_t K_t}{Z_t N_t} \right)^{\alpha-1} = \frac{\alpha}{\mu_t} \frac{Y_t}{K_t}, \quad (17)$$

$$\text{and } w_t = \frac{1-\alpha}{\mu_t} A_t Z_t \left(\frac{u_t K_t}{Z_t N_t} \right)^{\alpha} = \frac{1-\alpha}{\mu_t} \frac{Y_t}{N_t}. \quad (18)$$

The optimal utilization choice is given by

$$q_t [\delta_1 + \delta_2(u_t - 1)] = \frac{\alpha}{\mu_t} A_t K_t \left(\frac{u_t K_t}{Z_t N_t} \right)^{\alpha-1} = \frac{\alpha}{\mu_t} \frac{Y_t}{u_t}. \quad (19)$$

As a result, aggregate profits are $\Pi_t = \mu_t Y_t$. The logarithm of productivities A_t and Z_t evolve stochastically according to AR(1) processes

$$\ln A_t = (1 - \rho_A) \left(\ln A - \frac{\sigma_A^2}{2} \right) + \rho_A \ln A_{t-1} + \epsilon_t^A \quad \text{with } \epsilon_t^A \sim N(0, \sigma_A^2), \quad (20)$$

$$\text{and } \ln Z_t = (1 - \rho_Z) \left(\ln Z - \frac{\sigma_Z^2}{2} \right) + \rho_Z \ln Z_{t-1} + \epsilon_t^Z \quad \text{with } \epsilon_t^Z \sim N(0, \sigma_Z^2). \quad (21)$$

ρ_i and σ_i^2 with $i \in \{A, Z\}$ denote the autocorrelation of the log-technology shocks and the variance of their normally distributed innovations, while A and Z denote the unconditional means of the stochastic processes. Moreover, we allow time-varying depreciation rates δ_{0t} , which evolves according to

$$\delta_{0t} = \delta_0 + \epsilon_t^\delta \quad \text{with } \epsilon_t^\delta \sim N(0, \sigma_\delta^2). \quad (22)$$

Finally, capital goods producers take the relative price of capital goods, q_t , as given when determining their output. They face capital adjustment costs as in [Hayashi \(1982\)](#) and

maximize

$$\max_{\{I_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_{0,t} \left\{ q_t \left[I_t - \frac{\phi_t}{\kappa} \left(\frac{I_t}{K_t} - \delta_{0t} \right)^{\kappa} K_t \right] - I_t \right\}. \quad (23)$$

where $\kappa > 1$. To enhance the complexity of the model, the adjustment costs feature a non-linear element. In particular, ϕ_t is state-dependent and depends on the level of aggregate capital:

$$\phi_t = \begin{cases} \bar{\phi} & \text{if } K_t > \bar{K} \\ \underline{\phi} & \text{if } K_t \leq \bar{K} \end{cases} \quad (24)$$

where $\bar{K} \geq \underline{K} \geq 0$. Note that capital good producers take the adjustment costs as given as they depend on aggregate capital.

Optimization yields the optimality condition

$$q_t = \left[1 - \phi_t \left(\frac{I_t}{K_t} - \delta_{0t} \right)^{\kappa-1} \right]^{-1}. \quad (25)$$

Each capital goods producer will adjust its production, until (25) is satisfied. Since all capital goods producers are symmetric, we obtain a law of motion for aggregate capital

$$K_{t+1} = (1 - \delta(u_t)) K_t + I_t - \frac{\phi_t}{\kappa} \left(\frac{I_t}{K_t} - \delta \right)^{\kappa} K_t. \quad (26)$$

Having specified the production sector, we now describe the households in the economy.

Household sector: There exists a unit continuum of (potentially heterogeneous) households indexed by $i \in [0, 1]$ which maximize their lifetime utility discounted by the factor β . The households obtain utility from consumption c_{it} and disutility from supplying labor n_{it} . To smooth consumption, households accumulate capital k_{it+1} . The household's objective function is

$$U_{it} = \max_{c_{it}, n_{it}, k_{it+1}} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \zeta_t u(c_{it}, n_{it}), \quad (27)$$

with $\mathbb{E}_t$ denoting the expectation operator over all stochastic processes given the information set as of time t and $u(c_{it}, n_{it})$ denotes the per period felicity function of the household over consumption c_{it} and labor n_{it} . ζ_t is a stochastic aggregate shock to the discount factor in period t . The logarithm of the discount factor shock ζ_t evolves stochastically according

to an AR(1) process

$$\ln \zeta_t = -(1 - \rho_\zeta) \frac{\sigma_\zeta^2}{2} + \rho_\zeta \ln \zeta_{t-1} + \epsilon_t^\zeta \quad \text{with} \quad \epsilon_t^\zeta \sim N(0, \sigma_\zeta^2). \quad (28)$$

ρ_ζ denotes the autocorrelation of the logarithmic discount factor shock, and the shock ϵ_t^ζ is normally distributed with mean zero and variance σ_ζ^2 . Households optimize the objective function (27) subject to the budget constraint

$$(1 + \tau^C) c_{it} + q_t k_{it+1} = (q_t + (1 - \tau^K) r_t) k_{it} + (1 - \tau^L) w_t h_{it} n_{it} + T_t + \Pi_{it}. \quad (29)$$

r_t and w_t denote the interest rate and wage rate as specified above and h_{it} denotes households' idiosyncratic income component. τ^C , τ^K , and τ^L denote the value-added-tax, the capital income tax, and the labor income tax, while T_t denotes the transfers the households obtain from the government. Π_{it} denotes the individual part in aggregate profits. In all applications, households face a borrowing constraint, such that they are prohibited from holding negative amounts of assets. Individual productivity h_{it} evolves according to

$$\log h_{it} = -(1 - \rho_h) \frac{\sigma_h^2}{2} + \rho_h \log h_{it-1} + \epsilon_{it}^h \quad \text{with} \quad \epsilon_{it}^h \sim N(0, \sigma_h^2). \quad (30)$$

with ϵ_{it}^h as a normally distributed shock with variance σ_h^2 and mean zero.

The solution of the household problem can be characterized by the Euler equation on capital and the optimal labor supply schedule below

$$q_t u_C(c_{it}, n_{it}) = \beta \mathbb{E}_t \left[\frac{\zeta_{t+1}}{\zeta_t} (q_{t+1} + (1 - \tau^K) r_{t+1}) u_C(c_{it+1}, n_{it+1}) \right] \quad (31)$$

$$-u_L(c_{it}, n_{it}) = (1 - \tau^L) w_t h_{it} \frac{u_C(c_{it}, n_{it})}{1 + \tau^C}. \quad (32)$$

$u_C(c_{it}, n_{it}) = \frac{\partial u}{\partial c_{it}}(c_{it}, n_{it})$ denotes the partial derivative of the felicity function with respect to consumption and $u_L(c_{it}, n_{it}) = \frac{\partial u}{\partial n_{it}}(c_{it}, n_{it})$ denotes the partial derivative of the felicity function with respect to labor.

Government sector: The government levies distortionary capital and labor income taxation at flat rates τ^K and τ^L , and claims a value-added-tax τ^C on consumption. It uses the tax revenues to finance lump-sum transfers T_t to the household. Therefore, the role of the government is purely to redistribute between factor incomes and consumption and leisure. The budget constraint is

$$T_t = \tau^C C_t + \tau^K r_t K_t + \tau^L w_t N_t. \quad (33)$$

Government transfers T_t adjust residually to make the government budget constraint hold.

Market clearing and equilibrium: The labor market, the capital market, and the goods market have to clear at all periods. Labor and capital market clearing requires

$$N_t = \int_0^1 n_{it} di \quad \text{and} \quad K_t = \int_0^1 k_{it} di. \quad (34)$$

Given these aggregate quantities, prices are determined by their marginal products on the factor inputs as denoted in equations (17) and (18). The goods market clears when

$$Y_t = C_t + I_t, \quad (35)$$

where $I_t = K_{t+1} - (1 - \delta(u_t))K_t + \frac{\phi}{\kappa} \left(\frac{I_t}{K_t} - \delta_0 \right)^\kappa K_t$ denotes aggregate investment into next periods capital stock net of adjustment costs and $C_t = \int_0^1 c_{it} di$ is aggregate consumption. The goods market clears due to Walras-Law whenever the capital and the labor market clear.

Dynamic equilibrium: We define a dynamic equilibrium in this economy as follows. Firms and households take prices as given. Households behave optimally to maximize their lifetime utility (27) subject to the associated budget constraint (29) and the stochastic processes. Firms choose their factor inputs to maximize profits given their Cobb-Douglas production technology until the optimality conditions (17), (18), and (19). Lump sum taxes adjust such that the government budget constraint (33) holds, while the labor and asset markets (34), and the goods market (35) clear.

4 Proof of Concept

This section demonstrates generative economic modeling by applying our methodology to a range of models and comparing the results against those obtained with traditional global solution methods.

4.1 Generative Economic Modeling with an Analytical Solution

In this section, we illustrate our methodology using a stochastic representative agent version of the model, which admits an analytical solution following the approach of Brock and Mirman (1972). To solve the model analytically, we abstract from shocks to the discount

factor (ζ_t) and depreciation rate (δ_t). We keep the shocks to technology (A_t), productivity (Z_t), shocks to the markup (μ_t). Finally, we abstract from capital income taxation ($\tau^K = 0$) and assume full depreciation ($\delta_{0t} = 1$).⁹ Moreover, we abstract from household heterogeneity and let households be ex-ante identical by assuming away differences in idiosyncratic income $h_{i0} = 1$ and $\Pi_{it} = \Pi_t \forall i$ and initial capital holdings are identical $k_{i0} = K_0 \forall i$. We make households ex-post identical by disregarding idiosyncratic income risk $\sigma_h^2 = 0$. The absence of ex-ante or ex-post heterogeneity enables us to represent the household side through a representative agent. Therefore, we drop the individual index i to describe the variables of interest.

The following proposition combines the ingredients above to obtain an analytical solution to the model.

Proposition 1. *If all households are ex-ante and ex-post identical, depreciation is deterministic and full, $\delta(u_t) = 1$, capacity utilization is fixed at $u_t = 1$, there are no capital adjustment costs $\phi = 0$, the discount factor shock is inactive $\sigma_\zeta^2 = 0$, and per period felicity is of [King, Plosser and Rebelo \(1988\)](#) (KPR)-form given by $u(C_t, N_t) = \ln C_t - \omega \frac{N_t^{1+\gamma}}{1+\gamma}$. Then the policy functions of the representative household are*

$$N_t(\mu_t) = \left[\frac{\mu(1 - \tau^L)(1 - \alpha)}{\mu_t \omega (1 + \tau^C)(\mu - (1 - \tau^K)\alpha\beta)} \right]^{\frac{1}{1+\gamma}}, \quad (36)$$

$$C_t(K_t, A_t, Z_t, \mu_t) = \left(1 - \frac{\alpha\beta}{\mu_t} \right) Y_t(K_t, A_t, Z_t, N_t(\mu_t)), \quad (37)$$

$$\text{and } K_{t+1}(K_t, A_t, Z_t, \mu_t) = \frac{\alpha\beta}{\mu_t} Y_t(K_t, A_t, Z_t, N_t(\mu_t)), \quad (38)$$

where $Y_t(K_t, A_t, Z_t, N_t(\mu_t))$ denotes the Cobb-Douglas production function. Given the policy functions of the household, the prices in the economy can be determined by equations (17) and (18). The transfers to the households are then determined by the government budget constraint (33).

Proof. See Appendix I. □

Proposition 1 presents the solution for the representative agent economy. Fluctuations in the markup μ_t drive changes in labor supply N_t , while shocks to productivity (A_t, Z_t) affect output Y_t . Consumption C_t and capital investment K_{t+1} are linear functions of output. As

⁹ The combination of the assumptions renders the model unrealistic, as already noted by [Brock and Mirman \(1972\)](#) themselves. We do not employ the model for realistic reasons, but because it provides us with an analytical benchmark we can use. This also motivates the choice of our shocks.

Table 1 Parameter values of the analytical [Brock and Mirman \(1972\)](#) model

Parameter	Value	Description	Parameter	Value	Description
Households			Exogenous processes		
β	0.96	Discount factor	A	1.0	Steady state TFP
γ	5	Inverse Frisch	ρ_a	0.9	TFP persistence
ω	1.0	Scale labor disutility	σ_a	{0.0, 0.01, 0.05}	TFP std.
Firms			Z	1.0	Steady state labor prod.
α	0.33	Capital share	ρ_z	0.9	Labor prod. persistence
δ	1.0	Depreciation rate	σ_z	{0.0, 0.01, 0.05}	Labor prod. std.
Government			μ	1.5	Steady State Markup
τ^L	0.0%	Labor tax rate level	ρ_μ	0.9	Markup persistence
τ^R	0.0%	Capital tax rate level	σ_μ	{0.0, 0.01, 0.05}	Markup std.
τ^C	0.0%	Value-added tax rate level			

NOTE - All parameters in the table are calibrated to a yearly frequency.

noted by [Brock and Mirman \(1972\)](#), this model is highly simplistic and omits key real-world dynamics. Despite these limitations, we use it as a benchmark, providing an exact reference to evaluate our methodology.

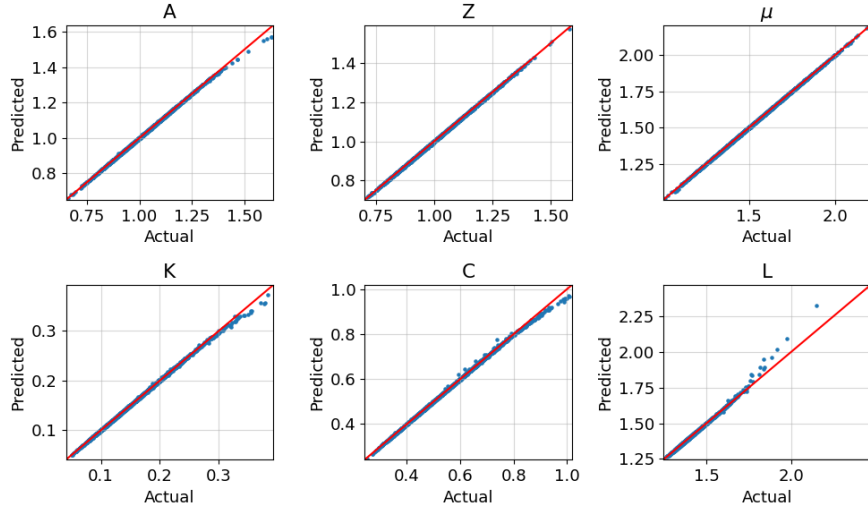
Table 1 illustrates the parameter values that we use for solving the model. We largely use standard values from the literature, but some variables require further explanation. First, we have a steady-state markup μ of 1.5, which is very high. We introduce such a high markup value such as to amplify the effects of markup shocks on the economy. Moreover, we shut down the government by setting all tax rates equal to zero. Finally, we do not only simulate the economy with fixed volatilities of the shocks but allow for different volatility levels. While the model only features three shocks, the shocks can have different volatilities, such that we generate data sets for all three shock combinations with different volatilities. This also challenges the surrogate network since it needs to learn the model dynamics for different shock volatilities.

Figure 2 illustrates the performance of our methodology. Panel 2a) compares the surrogate model's predictions to the actual data generated by the full model, demonstrating the accuracy with which our approach captures the true dynamics. Panel 2b) presents the errors of the surrogate model relative to the true data generation process, alongside the errors of a deep neural network trained directly on that process. This comparison allows us to assess the performance loss incurred by restricting our neural network to training only on satellite model data.

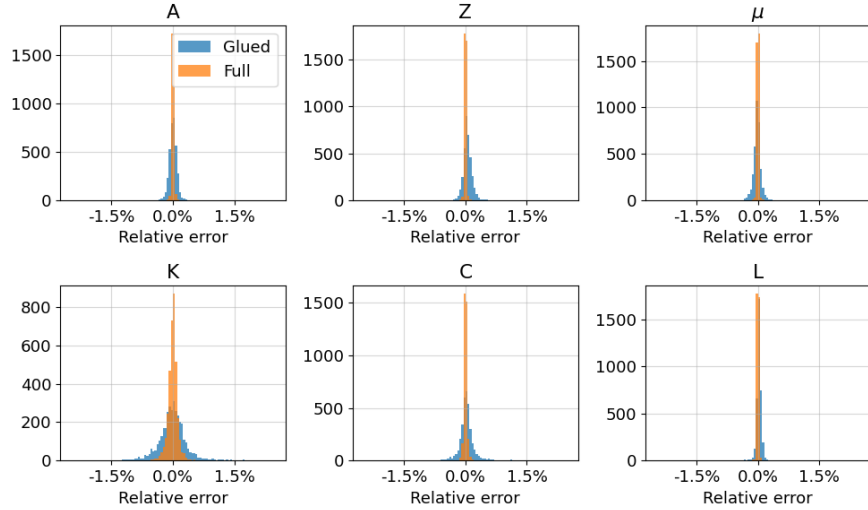
When evaluating our generative economic modeling algorithm against the true data in Panel 2a), we find that the methodology effectively captures the dynamics of the exogenous stochastic processes A_t , Z_t , and μ_t . The surrogate model performs well overall, with minor imprecision occurring only in the case of large and infrequent shocks, where it tends to underpredict values. For the endogenous variables K_t , C_t , and L_t , the model maintains

Figure 2 Evaluation of the surrogate model with an analytical RBC model

(a) Prediction of surrogate against data



(b) Error distribution of surrogate model



strong performance, albeit with slightly lower accuracy than for the exogenous processes. Specifically, it underpredicts capital K_t and consumption C_t at the upper end of the data range while slightly overpredicting labor supply L_t . This behavior likely stems from the limited number of observations in these regions, requiring the neural network to extrapolate. The fit can be further improved by expanding the dataset and optimizing the number of training epochs, enhancing the model's ability to capture extreme values more accurately.

We can assess the prediction errors of the neural network in relative terms in Panel 2b). The blue bins represent the relative errors of the surrogate model compared to the predicted variable values, while the orange bins show the corresponding errors of a deep

neural network trained on the full dataset rather than on slices generated from satellite models. Since the surrogate model is trained on limited data, it is expected that its relative errors are larger than those of the fully trained deep neural network.

Both neural networks achieve high accuracy in predicting the exogenous processes A_t , Z_t , and μ_t , with relative errors well below one percent. The largest relative errors for the surrogate model occur in the prediction of capital K_t , though they remain under one percent. Notably, even the deep neural network trained on the full dataset struggles to predict capital stock with high precision, highlighting an inherent challenge in predicting this variable. In contrast, the surrogate model performs exceptionally well in predicting consumption C_t and labor supply L_t , achieving a level of accuracy comparable to the fully trained neural network.

While the surrogate model exhibits some errors, its overall performance is strong. It effectively captures the dynamics of the full model, with only minor difficulties in predicting the capital stock—a challenge that even the fully trained neural network faces. Importantly, all errors remain small in relative terms, such that the method of generative economic modeling provides a reliable and useful approximation of the full model's dynamics.

4.2 Generative Economic Modeling with Global Solutions

In this section, we illustrate our methodology using a nonlinear medium-sized version of our RBC model. We abstract from shocks to labor-augmenting productivity Z_t and to the markup μ_t , hence shocks to TFP A_t are the only drivers of supply-side fluctuations. We assume that households remain ex-ante and ex-post identical, such that we can represent the household side with a representative agent. Moreover, households have KPR-preferences

$$u(C_t, N_t) = \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \omega \frac{N_t^{1+\gamma}}{1+\gamma} \quad (39)$$

such that optimal household behavior can be described by the Euler equation and the optimal labor supply condition:

$$\begin{aligned} C_t^{-\sigma} &= \beta \mathbb{E}_t [(q_{t+1} + (1 - \tau^K)r_{t+1})C_{t+1}^{-\sigma}] \\ \omega N_t^\gamma &= \frac{1 - \tau_t^L}{1 + \tau^C} w_t C_t^{-\sigma}. \end{aligned}$$

Contrary to the former section, we allow for capacity utilization choice and allow for capital adjustment costs with a nonlinear specification as illustrated in equations (23) and (24). The model is solved with global methods, specifically policy function iteration, to

Table 2 Parameter values of the nonlinear medium-sized RBC model

Parameter	Value	Description	Parameter	Value	Description
Households			Firms		
σ	1	Risk aversion	α	0.33	Capital share
β	0.99	Discount factor	δ_0	0.025	Depreciation rate
γ	1	Inverse Frisch	δ_1	αY^{SS}	Depreciation rate
ω	0.5	Scale labor disutility	δ_2	$5\delta_1$	Depreciation rate
Exogenous processes			κ	2	Cap. adj. cost curvature
A	1.0	Steady state TFP	$\bar{\phi}$	2.5	High slope of cap. adj. cost
ρ_a	0.95	TFP persistence	$\underline{\phi}$	0.025	Low slope of cap. adj. cost
σ_a	0.01	TFP std.	Government		
ρ_ζ	0.95	Discount factor persistence	τ^L	0.0%	Labor tax rate level
σ_ζ	0.05	Discount factor std.	τ^R	0.0%	Capital tax rate level
σ_δ	0.004	Depreciation std.	τ^C	0.0%	Value-added tax rate level

NOTE - All parameters in the table are calibrated to a quarterly frequency.

account for all nonlinear features. Within the class of policy function iteration methods, we use time iteration with linear interpolation as in [Richter, Throckmorton and Walker \(2014\)](#) and [Bianchi, Melosi and Rottner \(2021\)](#).

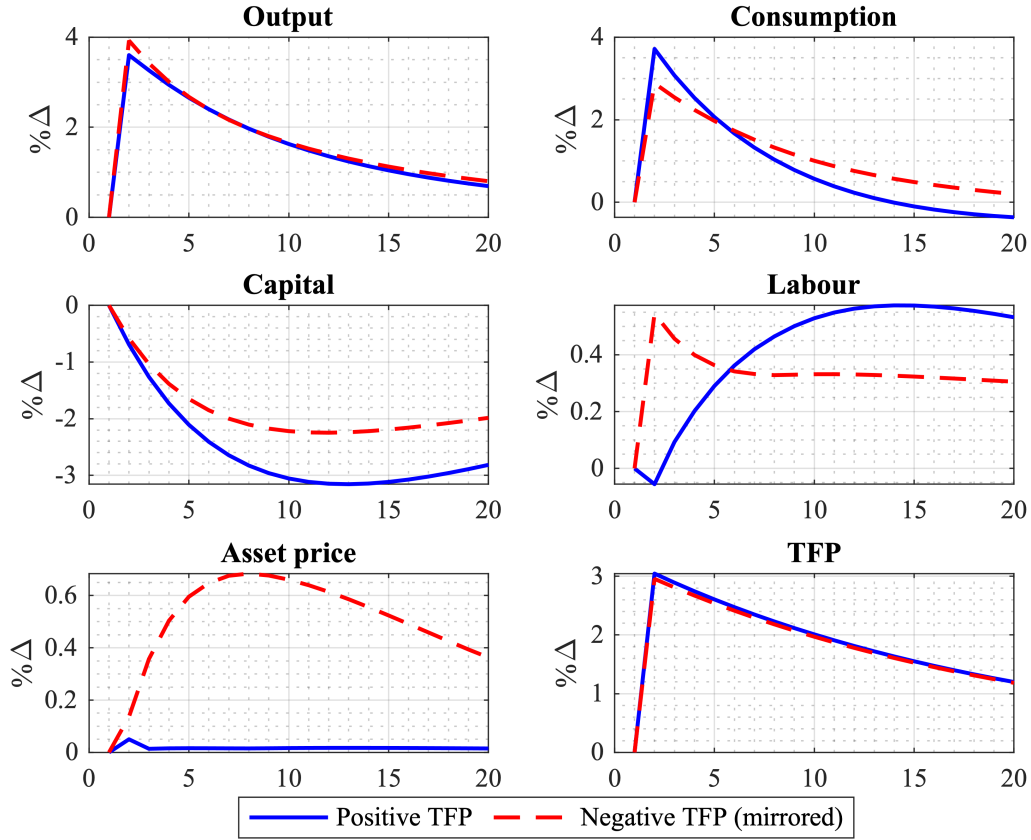
Table 2 presents the parameter choices for solving the model. Most parameters align with standard values in the literature but are calibrated at a quarterly frequency. Compared to the model in Section 3.1, we introduce partial depreciation, a capacity utilization choice, a lower Frisch elasticity of labor supply, and capital adjustment costs. Additionally, we incorporate nonlinear capital adjustment costs by allowing ϕ_t to take values $\bar{\phi}$ and $\underline{\phi}$ depending on the capital stock. The other parameters are standard.

The model features strong nonlinearities as the impulse response functions (IRFs) in Figure 3 highlight. In this simulation, we compare the impact of an expansionary and contractionary three-standard deviation TFP shock. We display the percentage deviation from the stochastic steady state and mirror the IRFs of the negative TFP for easier comparison. The state-dependent investment costs result in strong differences between positive and negative shocks. We observe a similar behavior when evaluating the preference and discount rate shock. This is an important precondition for our analysis as we want to evaluate the performance of our approach in a highly nonlinear environment.

Figure 4 illustrates the performance of our algorithm for the medium-sized nonlinear DSGE model. Panel 4a) illustrates the fit of the surrogate model against the true data, while panel 4b) illustrates the relative errors of the surrogate model and of a deep neural network that is trained on the dataset generated by the full model.

Comparing the surrogate model's predictions with the true data in panel 4a) demonstrates that our methodology provides a strong fit, even for a medium-sized DSGE model with nonlinearities. The predicted values closely align with the true values, addressing the

Figure 3 IRFs and Nonlinear Propagation of the TFP Shock

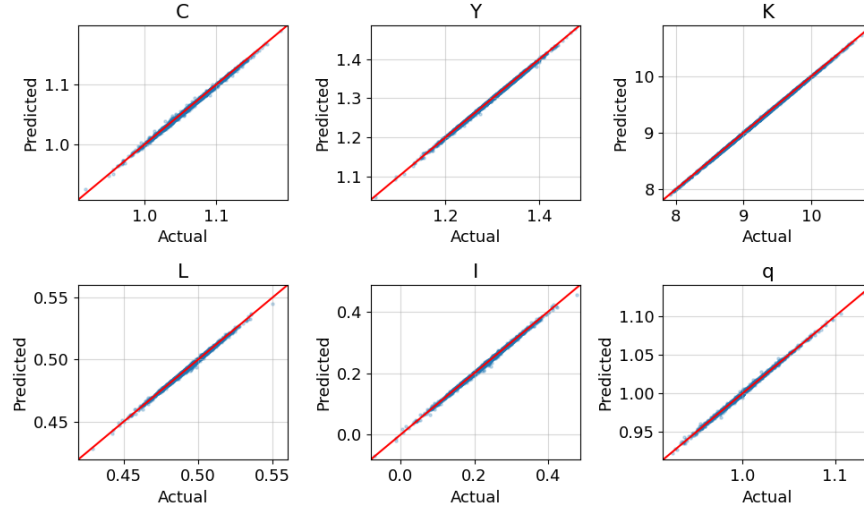


issue of under- and overprediction observed in the analytical version from the previous section. The model performs well in predicting exogenous processes (not illustrated) and provides a reasonable fit for endogenous variables. Among these, capital K_t exhibits the best fit, while investment I_t shows the largest deviations due to its high volatility, driven by nonlinear adjustment costs. Notably, the model's ability to predict the price of capital q_t remains unaffected by these nonlinearities.

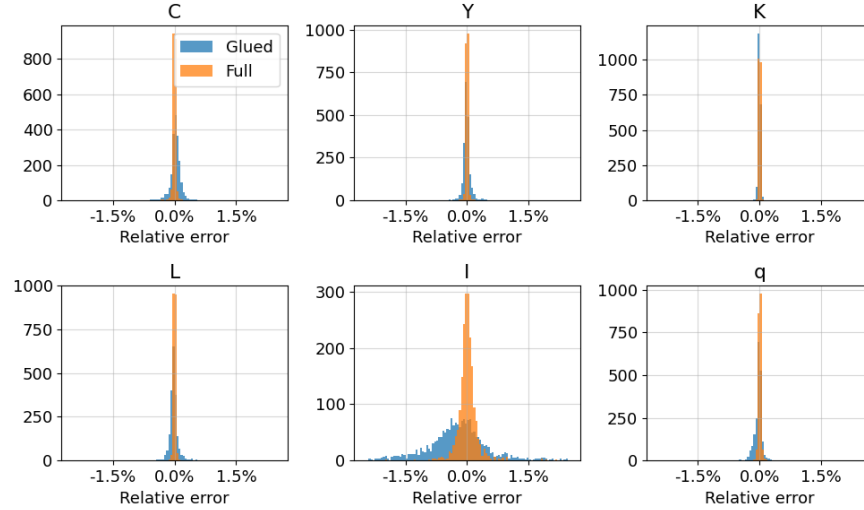
The relative prediction errors in panel 4b) indicate that the surrogate model effectively captures the true dynamics of the approximated model. For all endogenous variables except investment I_t , relative errors remain well below one percent and are comparable to those generated by the deep neural network trained on the full dataset. In contrast, the relative errors for I_t reach up to 2%, highlighting a weaker performance in predicting investment. Furthermore, the deep neural network trained on the full dataset significantly outperforms the surrogate model. A possible explanation for this discrepancy is the nonlinear dynamics underlying investment: certain shocks have minimal impact on I_t , limiting the information available in the satellite datasets. As a result, the surrogate model extracts less relevant

Figure 4 Evaluation of the surrogate model with a nonlinear medium-sized RBC model

(a) Prediction of surrogate against data



(b) Error distribution of surrogate model



information during training, reducing its predictive accuracy for investment.

Overall, the surrogate model effectively captures the dynamics of the underlying model and even improves its forecasting performance for endogenous variables compared to the previous section. Although forecast errors are higher for variables exhibiting nonlinearities, such as investment I_t , the relative errors remain small.

4.3 Generative Economic Modeling with Heterogeneous Agents

This section illustrates the application of our methodology to a model with heterogeneous households. Contrary to the former sections, we allow for ex-ante and ex-post heterogeneity, as in [Krusell and Smith \(1998\)](#). Consequently, the joint distribution of wealth and income states becomes a state variable. For now, we abstract from a capacity utilization choice $u_t = 1$ and capital adjustment costs $\phi_t = \kappa = 0$. Households maximize the simple CRRA-felicity function

$$u(c_{it}) = \frac{c_{it}^{1-\sigma} - 1}{1-\sigma}$$

subject to the budget constraint in equation (29). The absence of disutility from labor implies that households always supply one unit of labor. We shut down the government with the exception of labor income taxation. Employed households pay a linear tax on their labor income such that the unemployed households obtain τ times the labor income of employed households as unemployment benefits.

We allow for three aggregate shocks (A_t, ζ_t, δ_t) . To solve the model with heterogeneous agents and three aggregate shocks we discretize the aggregate state space. We follow [Busch and Ludwig \(2024\)](#) in modeling TFP A_t as a two-state Markov chain representing the US business cycle. The discount factor shock ζ_t and depreciation rate shock δ_t are also discretized using two-state Markov chains. For now, we assume that the Markov chains governing ζ_t and δ_t are independent of TFP and idiosyncratic risk.

Idiosyncratic income shocks h_{it} follow a two-state process, where the high state represents employment and the low state represents unemployment. As in the original paper, the transition matrix for idiosyncratic income risk depends on aggregate productivity, making unemployment duration countercyclical: households remain unemployed for an average of 1.5 periods in good states and 2.5 periods in bad states. The unemployment rate also varies with the cycle, reaching 4% in good states and 10% in bad states. We impose the conditions $\frac{\pi_{gb00}}{\pi_{gb}} = 1.25 \frac{\pi_{bb00}}{\pi_{bb}}$ and $\frac{\pi_{bg00}}{\pi_{bg}} = 0.75 \frac{\pi_{gg00}}{\pi_{gg}}$, where $\pi_{aa'\epsilon\epsilon'}$ denotes the probability of transitioning to future aggregate state a' and idiosyncratic state ϵ' , given today's aggregate state a and idiosyncratic state ϵ , and $\pi_{aa'}$ represents the aggregate state transition matrix.

Table 3 illustrates the parameter values we use for the solution of the satellite models and the simulation of the data. We use largely standard values or the values used by [Krusell and Smith \(1998\)](#) themselves. With respect to the duration of the discount factor states and the depreciation states, we use calibrations that generate stationary distributions and do not induce explosive savings dynamics in the model with all shocks active simultaneously. The discount factor shock induces an additional savings motive for households, while the depreciation shock induces the opposite as in the next period households could be faced

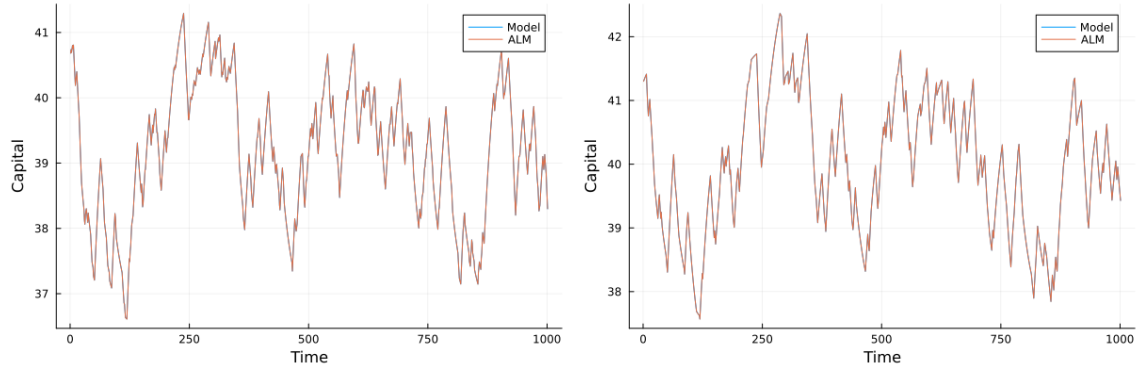
Table 3 Parameter values of the [Krusell and Smith \(1998\)](#) model

Parameter	Value	Description	Parameter	Value	Description
Households					
β	0.99	Discount factor	$D_{\zeta g}$	8	Dur. good disc. factor state
σ	1.0	Risk aversion	$D_{\zeta b}$	8	Dur. low disc. factor state
ζ	[0.99, 1.01]	Discount factor states	u_g	0.04	Unempl. rate in good state
			u_b	0.1	Unempl. rate in bad state
Firms					
α	0.36	Capital share	$D_{\delta g}$	8	Duration of good TFP state
A	[0.99, 1.01]	TFP states	$D_{\delta b}$	8	Duration of bad TFP state
δ	[0.0275, 0.025]	Depreciation rate states	$D_{\delta g}$	8	Duration good depr. state
			$D_{\delta b}$	2	Duration bad depr. state
Government					
τ	0.15	Unemployment benefits	τ^R	0.0%	Capital tax rate level
τ^L	$\frac{u_t}{1-u_t} \tau w_t$	Labor tax rate level	τ^C	0.0%	VAT rate level

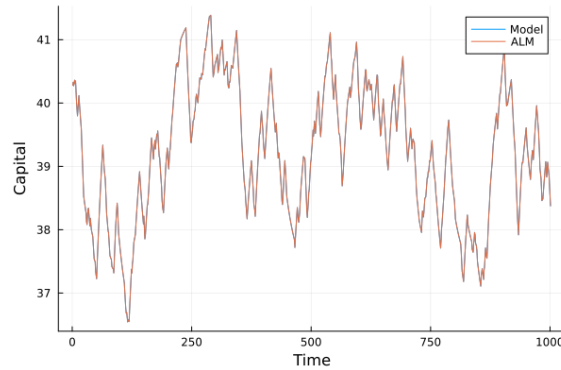
NOTE - All parameters in the table are calibrated to a quarterly frequency.

Figure 5 Model implied series of capital and perceived aggregate law of motion (ALM)

(a) Capital series with TFP and delta shocks (b) Capital series with TFP and beta shocks



(c) Capital series with TFP, beta and delta shocks



with an unexpectedly high depreciation rate.¹⁰

We provide a detailed description of the solution algorithm for the heterogeneous agent

¹⁰ Solving a model with variable depreciation is numerically very unstable since small infrequent fluctuations in the depreciation rate result in very precautionary savings behavior.

economy with multiple aggregate shocks in Appendix III. On the following, we only provide a brief overview about the solution method. To solve the household problem, we use the endogenous grid point method, originally developed by [Carroll \(2006\)](#). For aggregation and simulation of the economy, we use non-stochastic simulation of [Young \(2010\)](#). Since the model includes aggregate shocks, households require a perceived aggregate law of motion for capital to solve the consumption-savings problem. Following [Krusell and Smith \(1998\)](#), we specify a state-dependent log-linear perceived law of motion that households use for forecasting. To address the concerns raised by [Den Haan \(2010\)](#), we iteratively update this perceived law of motion until its coefficients converge and the gap between the true and perceived laws of motion for capital is small. Figure 5 illustrates the true law of motions of capital and the perceived law of motions for the satellite models and the full model. As the figure illustrates, the model-generated capital series and the perceived law of motion of capital are indistinguishable.

Preliminary results from our heterogenous agent economy agree with the results we obtain in the former sections. The predicted values closely align with this reference line across all variables, demonstrating a strong overall fit. However, the analysis in this section is still in progress. Some key results, particularly related to the application of our method, are not yet available. These will be included in a future version.

5 Conclusion

Our study introduces generative economic modeling, a novel approach that combines conventional solution methods with artificial intelligence to overcome computational barriers in solving complex dynamic economic models. By using neural networks trained on data generated from satellite models, we provide an alternative to standard deep learning-based approaches, which often require extensive fine-tuning and can suffer from instability due to their endogenous feedback loops. In contrast, our methodology ensures stability and scalability by using precomputed solutions from conventional methods, allowing for efficient training and accurate approximations of the full economic model.

The results demonstrate that this approach successfully captures model dynamics with high precision, yielding prediction errors comparable to those of deep neural networks trained on full model data. Importantly, our method extends the applicability of conventional global solution methods by using recent advances in artificial intelligence. This is particularly valuable for models featuring higher-order nonlinearities and heterogeneous agents, where the curse of dimensionality poses significant computational challenges.

Our general approach offers several promising avenues in the future. First, it can be

applied to more complex environments, such as solving nonlinear HANK models by training on simplified RANK and linearized HANK models. Second, it has the potential to enhance model estimation techniques, where fast and reliable solutions are critical. Lastly, we are developing evaluation metrics to assess neural network performance systematically, which will guide the optimal design of satellite models and network specifications.

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Appendix

I Appendix: Derivation of the analytical model

This section derives the proof related to the analytical model in section 4.1.

Proof. The proof employs a guess-and-verify approach. Guess that the policy function for savings is given by $K_{t+1} = \Gamma Y_t$. Substituting the guess into the goods market clearing condition (35) while imposing the parameter restriction $\delta = 1$ yields

$$C_t = (1 - \Gamma)Y_t.$$

We use the two guesses and substitute into the Euler equation (31)

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left[\frac{\frac{\alpha}{\mu_{t+1}} \frac{Y_{t+1}}{K_{t+1}}}{C_{t+1}} \right] \Leftrightarrow \frac{1}{(1 - \Gamma)Y_t} = \beta \mathbb{E}_t \left[\frac{\alpha}{\mu_{t+1}(1 - \Gamma)K_{t+1}} \right],$$

from which it is straightforward to see that $\Gamma = \frac{\alpha\beta}{\mu}$ given that $\mathbb{E}_t \frac{1}{\mu_{t+1}} = \mu^{-1}$ with $\rho_\mu = 0$. Note that the value-added-tax $(1 + \tau^C)$ drops from the Euler equation, since it is constant over time. To obtain the policy function (36) we substitute the guesses with specified Γ into the labor-supply condition (32)

$$\frac{(1 - \tau_t^N)w_t}{(1 + \tau^C)C_t} = \omega N_t^\gamma \Leftrightarrow \frac{(1 - \tau_t^N)(1 - \alpha)}{\mu_t(1 + \tau^C)(1 - \Gamma)N_t} = \omega N_t^\gamma,$$

from which we obtain expression (36) when solving for N_t . □

With KPR-preferences with log-felicity over consumption the income and substitution effect of wage changes cancel out. Therefore, only shocks to the wage tax rate τ_t^L and the markup μ_t impact the level of equilibrium labor supply.

II Appendix: Equilibrium and steady-state of the model

This section illustrates the steady state of the economy. We first illustrate all equations that characterize the equilibrium before characterizing the steady state.

Equilibrium characterization: In a symmetric equilibrium, the production side is char-

acterized by

$$Y_t : \quad Y_t = A_t(u_t K_t)^\alpha (Z_t N_t)^{1-\alpha} \quad (40)$$

$$r_t : \quad r_t + q_t \delta(u_t) = \frac{\alpha}{\mu_t} \frac{Y_t}{K_t} \quad (41)$$

$$w_t : \quad w_t = \frac{1-\alpha}{\mu_t} \frac{Y_t}{N_t} \quad (42)$$

$$u_t : \quad q_t[\delta_1 + \delta_2(u_t - 1)] = \frac{\alpha}{\mu_t} \frac{Y_t}{u_t} \quad (43)$$

$$\Pi_t : \quad \Pi_t = \mu_t Y_t \quad (44)$$

$$q_t : \quad q_t = \left[1 - \phi \left(\frac{I_t}{K_t} - \delta_{0t} \right)^{\kappa-1} \right]^{-1} \quad (45)$$

$$I_t : \quad K_{t+1} = (1 - \delta(u_t)) K_t + I_t - \frac{\phi}{\kappa} \left(\frac{I_t}{K_t} - \delta_{0t} \right)^\kappa K_t \quad (46)$$

With four exogenous processes:

$$\mu_t : \quad \ln \mu_t = (1 - \rho_\mu) \left(\ln \mu - \frac{\sigma_\mu^2}{2} \right) + \rho_\mu \ln \mu_{t-1} + \epsilon_t^\mu \quad \text{with} \quad \epsilon_t^\mu \sim N(0, \sigma_\mu^2) \quad (47)$$

$$A_t : \quad \ln A_t = (1 - \rho_A) \left(\ln A - \frac{\sigma_A^2}{2} \right) + \rho_A \ln A_{t-1} + \epsilon_t^A \quad \text{with} \quad \epsilon_t^A \sim N(0, \sigma_A^2), \quad (48)$$

$$Z_t : \quad \ln Z_t = (1 - \rho_Z) \left(\ln Z - \frac{\sigma_Z^2}{2} \right) + \rho_Z \ln Z_{t-1} + \epsilon_t^Z \quad \text{with} \quad \epsilon_t^Z \sim N(0, \sigma_Z^2), \quad (49)$$

$$\delta_{0t} : \quad \delta_{0t} = \delta_0 + \epsilon_t^\delta, \quad (50)$$

The household side is characterized by the aggregate functions $\mathcal{C}_t$, $\mathcal{N}_t$ and $\mathcal{K}_{t+1}$ that represent the aggregate consumption, labor, and capital function of the (potentially heterogeneous) household side of the model. We can characterize the aggregate functions as

$$\mathcal{C}_t : \quad \mathcal{C}_t(\{r_t, w_t, q_t, \zeta_t, \tau^k, \tau^L, \tau^C\}) = \int c_{it}^* d\Theta_t, \quad (51)$$

$$\mathcal{N}_t : \quad \mathcal{N}_t(\{r_t, w_t, q_t, \zeta_t, \tau^k, \tau^L, \tau^C\}) = \int n_{it}^* d\Theta_t, \quad (52)$$

$$\mathcal{K}_{t+1} : \quad \mathcal{K}_{t+1}(\{r_t, w_t, q_t, \zeta_t, \tau^k, \tau^L, \tau^C\}) = \int k_{it}^* d\Theta_t, \quad (53)$$

with the integral integrating over the distribution of households Θ_t and lowercase letters with asterisks denoting the policy functions of a household in period t that are determined

by the first-order conditions

$$c_{it} : \quad q_t u_C(c_{it}, n_{it}) = \beta \mathbb{E}_t \left[\frac{\zeta_{t+1}}{\zeta_t} (q_{t+1} + (1 - \tau^K) r_{t+1}) u_C(c_{it+1}, n_{it+1}) \right] \quad (54)$$

$$n_{it} : \quad -u_L(c_{it}, n_{it}) = (1 - \tau^L) w_t h_{it} \frac{u_C(c_{it}, n_{it})}{1 + \tau^C}, \quad (55)$$

and residually via the budget constraint:

$$k_{it+1} : (1 + \tau^C) c_{it} + q_t k_{it+1} = (q_t + (1 - \tau^K) r_t) k_{it} + (1 - \tau^L) w_t h_{it} n_{it} + T_t + \Pi_{it} \quad (56)$$

Generally, the distribution Θ_t of agents develops according to a law of motion

$$\Theta_{t+1} : \quad \Theta_{t+1}(h_{it+1}, k_{it+1}) = \int_{k_{it+1}=k_{it}^*} \Phi(h_{it}, h_{it+1}) d\Theta_t(h_{it}, k_{it}), \quad (57)$$

with Φ denoting the transition of idiosyncratic states. In the representative agent case, the aggregate functions just boil down to the first-order conditions (54), (55), and (56). On the household side, we then have the exogenous processes:

$$\zeta_t : \quad \ln \zeta_t = -(1 - \rho_\zeta) \frac{\sigma_\zeta^2}{2} + \rho_\zeta \ln \zeta_{t-1} + \epsilon_t^\zeta \quad \text{with} \quad \epsilon_t^\zeta \sim N(0, \sigma_\zeta^2). \quad (58)$$

$$h_{it} : \quad \log h_{it} = -(1 - \rho_h) \frac{\sigma_h^2}{2} + \rho_h \log h_{it-1} + \epsilon_{it}. \quad (59)$$

Finally, the government sector is characterized by

$$T_t : \quad T_t = \tau^C C_t + \tau^K r_t K_t + \tau^L w_t N_t, \quad (60)$$

while market clearing is characterized by

$$N_t = \mathcal{N}_t, \quad (61)$$

$$K_t = \mathcal{K}_t, \quad (62)$$

$$C_t = \mathcal{C}_t \quad (63)$$

$$Y_t = C_t + I_t, \quad (64)$$

Hence, in equilibrium, the 25 equations above determine the 25 variables $\{Y_t, r_t, w_t, u_t, \Pi_t, q_t, I_t, \mu_t, A_t, Z_t, \delta_{0t}, \{c_{it}^*\}_i, C_t, \mathcal{C}_t, \{n_{it}^*\}_i, N_t, \mathcal{N}_t, K_t, \{k_{it}^*\}_i, K_{t+1}, \mathcal{K}_{t+1}, \zeta_t, h_{it}, T_t, \Theta_t\}$.

Steady-State: Based on the description of the equilibrium, we now can turn to describe the steady state. First, except for idiosyncratic income risk h_{it} all exogenous processes

revert to their long-run averages such that:

$$\mu^{SS} : \quad \ln \mu^{SS} = \ln \mu - \frac{\sigma_\mu^2}{2} \quad (65)$$

$$A^{SS} : \quad \ln A^{SS} = \ln A - \frac{\sigma_A^2}{2} \quad (66)$$

$$Z^{SS} : \quad \ln Z^{SS} = \ln Z - \frac{\sigma_Z^2}{2} \quad (67)$$

$$\delta_0^{SS} : \quad \delta_0^{SS} = \delta_0 \quad (68)$$

$$\zeta^{SS} : \quad \ln \zeta^{SS} = -\frac{\sigma_\zeta^2}{2}. \quad (69)$$

In the heterogeneous agent version of the model in the steady state, there still exists heterogeneity on the household side in a steady state. The steady state on the household side is then characterized by constant *aggregates*, while still featuring fluctuations at the individual household level. Hence

$$\mathcal{C}^{SS} : \quad \mathcal{C}^{SS} = \mathcal{C}_t = \mathcal{C}_{t+1} = \int c_i^* d\Theta^{SS}, \quad (70)$$

$$\mathcal{N}^{SS} : \quad \mathcal{N}^{SS} = \mathcal{N}_t = \mathcal{N}_{t+1} = \int n_i^* d\Theta^{SS}, \quad (71)$$

$$\mathcal{K}^{SS} : \quad \mathcal{K}^{SS} = \mathcal{K}_{t+1} = \mathcal{K}_{t+2} = \int k_i^* d\Theta^{SS}, \quad (72)$$

with Θ^{SS} denoting the time-invariant stationary distribution of households that solves

$$\Theta^{SS} : \quad \Theta^{SS}(h_i, k_i) = \int_{k_i=k_i^*} \Phi(h_{it}, h_{it+1}) d\Theta^{SS}(h_i, k_i). \quad (73)$$

The lower-case letters with asterisks subscript i denote the policy functions given constant aggregates. In the representative version of the model, this boils down to

$$r^{SS} : \quad q^{SS} = \beta [(q^{SS} + (1 - \tau^K)r^{SS})] \quad (74)$$

$$\mathcal{N}^{SS} : \quad -u_L(\mathcal{C}^{SS}, \mathcal{N}^{SS}) = \frac{1 - \tau^L}{1 + \tau^C} w^{SS} u_C(\mathcal{C}^{SS}, \mathcal{N}^{SS}), \quad (75)$$

$$\mathcal{C}^{SS} : \quad (1 + \tau^C)\mathcal{C}^{SS} = (1 - \tau^L)w^{SS}\mathcal{N}^{SS} + (1 - \tau^K)r^{SS}\mathcal{K}^{SS} + T^{SS} + \Pi^{SS}. \quad (76)$$

Using the market clearing conditions determines aggregate consumption and aggregate labor

$$\mathcal{C}^{SS} = C^{SS} \quad \text{and} \quad \mathcal{N}^{SS} = N^{SS} \quad (77)$$

We use steady-state labor supply N^{SS} and the steady-state real interest rate r^{SS} to pin down the production side:

$$Y^{SS} : \quad Y^{SS} = A^{SS} (u^{SS} K^{SS})^\alpha (Z^{SS} N^{SS})^{1-\alpha} \quad (78)$$

$$K^{SS} : \quad r^{SS} + q^{SS} \delta(u^{SS}) = \frac{\alpha}{\mu^{SS}} \frac{Y^{SS}}{K^{SS}} \quad (79)$$

$$w^{SS} : \quad w^{SS} = \frac{1-\alpha}{\mu^{SS}} \frac{Y^{SS}}{N^{SS}} \quad (80)$$

$$u^{SS} : \quad q_t [\delta_1 + \delta_2 (u^{SS} - 1)] = \frac{\alpha}{\mu^{SS}} \frac{Y^{SS}}{u^{SS}} \quad (81)$$

$$\Pi^{SS} : \quad \Pi_t = \mu_t Y_t \quad (82)$$

$$q^{SS} : \quad q^{SS} = \left[1 - \phi \left(\frac{I^{SS}}{K^{SS}} - \delta_0^{SS} \right)^{\kappa-1} \right]^{-1} \quad (83)$$

$$I^{SS} : \quad I^{SS} = \left(\delta(u^{SS}) + \frac{\phi}{\kappa} \left(\frac{I^{SS}}{K^{SS}} - \delta_0^{SS} \right)^\kappa \right) K^{SS} \quad (84)$$

Government transfers are then determined by

$$T^{SS} : \quad T^{SS} = \tau^C C^{SS} + \tau^K r^{SS} K^{SS} + \tau^L w^{SS} N^{SS}, \quad (85)$$

Finally, the capital and the goods market have to clear

$$K^{SS} = \mathcal{K}^{SS}, \quad (86)$$

$$\text{and } Y^{SS} = C^{SS} + I^{SS}. \quad (87)$$

Having defined all equations that have to hold in steady state, we can now characterize the solution for a specific steady state.

Steady-State determination: It is easiest to start by imposing some normalization on the production side. We want to normalize capacity utilization u^{SS} and the price for capital to unity in the steady state q^{SS} . From equations (81) and (83) this derives the parameter restrictions

$$\frac{I^{SS}}{K^{SS}} = \delta_0^{SS} \quad \text{and} \quad \delta_1 = \frac{\alpha}{\mu^{SS}} Y^{SS}. \quad (88)$$

which pins down the depreciation values besides investment in steady state. The rest of the steady state can then be characterized by a joint solution of the household side and the firm side that satisfies market clearing conditions. In the heterogenous agent case, we solve the household problem for each household along the state space given a guess for aggregate

capital K_t and aggregate labor N_t , obtain the aggregate labor function (71) and capital function (72) and update the guesses for capital and labor until labor market clearing (77) and capital market clearing (86) are satisfied.

In the representative agent case, equation (74) determines the steady state real interest rate as

$$r^{SS} : \quad r^{SS} = \frac{1 - \beta}{\beta(1 - \tau^K)}. \quad (89)$$

In both cases, an interest rate also determines the steady-state capital-to-labor ratio and wage rate

$$K^{SS} : \quad \frac{K^{SS}}{N^{SS}} = \left(\frac{\alpha}{r^{SS} + \delta_0^{SS}} \frac{A^{SS}(Z^{SS})^{1-\alpha}}{\mu^{SS}} \right)^{\frac{1}{1-\alpha}} \quad (90)$$

$$w^{SS} : \quad w^{SS} = (1 - \alpha) \frac{A^{SS}(Z^{SS})^{1-\alpha}}{\mu^{SS}} \left(\frac{K^{SS}}{N^{SS}} \right)^\alpha. \quad (91)$$

III Appendix: Solving the heterogenous agent version

This section presents the solution to the heterogeneous agent model following the methodology of [Krusell and Smith \(1997\)](#) and [Krusell and Smith \(1998\)](#).¹¹ In the consumption-savings problem, households require a prediction of next period's capital given today's state space. With heterogeneous agents, this would typically require households to track the entire distribution of households over the state space, Θ_t as an additional state variable, which renders the problem numerically intractable. To address this, Krusell and Smith demonstrate that households do not need to keep track of the full distribution Θ_t , but a few moments of the distribution suffice to forecast future capital. Specifically, they approximate the law of motion for capital using its mean, allowing households to form expectations based on a simplified perceived law of motion. Following the original paper, we assume that the law of motion takes the following state-dependent functional form:

$$\ln K_{t+1} = \beta_0 + \beta_1 A_t + \beta_2 \zeta_t + \beta_3 \delta_t + \beta_4 \ln K_t + \beta_5 A_t \ln K_t + \beta_6 \zeta_t \ln K_t + \beta_7 \delta_t \ln K_t, \quad (92)$$

where we extend the original specification by including additional interaction terms. The law of motion (92) allows both the slope and the intercept to vary with the aggregate state, which is characterized by $\{A_t, \zeta_t, \delta_t\}$.

The solution algorithm consists of an inner and an outer loop. The outer loop iterates

¹¹ The code is available at <https://github.com/Fabio-Stohler/KS1998>.

until the coefficients in the regression equation (92) converge. Let

$$\beta^n = (\beta_0^n, \beta_1^n, \beta_2^n, \beta_3^n, \beta_4^n, \beta_5^n, \beta_6^n, \beta_7^n)'$$

denote the vector of regression coefficients for the perceived law of motion of iteration n of the algorithm. Convergence is determined by checking whether the coefficients remain unchanged across iterations. If the coefficients changed by less than a small ϵ , the algorithm terminates.¹²

The inner loop iterates until the household problem is globally solved for a given perceived law of motion for capital. To solve the household side, we discretize the space $(k_{it}, h_{it}, K_t, A_t, \zeta_t, \delta_t)$ and use the endogenous grid-point method (EGM) of [Carroll \(2006\)](#) to solve the household problem given the stochastic processes and the perceived law of motion. Household policies are updated iteratively until the (inverse) marginal values of consumption converge. Once the household problem is solved globally, we aggregate and simulate the economy for T periods using the stochastic simulation method of [Young \(2010\)](#). Finally, using the simulated time series of capital and the aggregate states, we estimate the regression equation (92) to update the law of motion.

We illustrate the algorithm as follows. Let $\vec{k}$, $\vec{h}$, $\vec{A}$, $\vec{\zeta}$, and $\vec{\delta}$ denote the discretized vectors of individual capital holdings, individual productivity, aggregate productivity, the aggregate discount factor, and aggregate depreciation, respectively.

1. For each realization of aggregate states $\{K_t, A_t, \zeta_t, \delta_t\}$ compute labor L_t (which depends on the aggregate state), as well as the interest rate r_t and the wage rate w_t as the marginal products of capital and labor. For each realization of the individual state space $\{k_{it}, h_{it}, K_t, A_t, \zeta_t, \delta_t\}$ compute household incomes.
2. Initialize the coefficients for the law of motion (92). Typically, the intercepts $(\beta_0, \beta_1, \beta_2, \beta_3)$ are set to zero, and the slopes $(\beta_4, \beta_5, \beta_6, \beta_7)$ to one.
3. Given the coefficients for the law of motion, solve the household problem using EGM¹³:

- (a) Initialize guesses for the policy functions c_{it}^0 , k_{it}^0 defined on the state space $(k_{it}, h_{it}, K_t, A_t, \zeta_t, \delta_t)$. Create an initial guess for the marginal value function

¹² We also verify that the true law of motion for capital closely matches the perceived law of motion, which is generally the case.

¹³ There exist numerous resources that go into detail in the illustration of the method. We only highlight the differences that occur due to the presence of aggregate risk. The interested reader might consult [Carroll \(2006\)](#), [Barillas and Fernández-Villaverde \(2007\)](#), [Hintermaier and Koeniger \(2010\)](#), and the appendix of [Bayer et al. \(2019\)](#).

$\frac{\partial V_{it}^0}{\partial k_{it}} = (1+r) \frac{\partial u(c_{it}^0)}{\partial c_{it}}$. The superscript denotes the iteration step mm of the EGM algorithm.

- (b) For each realization of the aggregate state today $\{K_t, A_t, \zeta_t, \delta_t\}$ forecast next period's capital stock using the perceived law of motion. Let $\tilde{K}_{t+1}$ denote the forecasted capital stock according to the perceived law of motion. Interpolate the marginal value $\frac{\partial V_{it}^{n-1}}{\partial k_{it}}$ from the exogenous grid $\vec{K}$ onto the perceived value in the next period $\tilde{K}_{t+1}$. Finally, compute the expected marginal value by integrating over the realizations of the aggregate $\{A_t, \zeta_t, \delta_t\}$ and idiosyncratic states $\{h_{it}\}$ and discount the expected value with the discount factor.¹⁴
 - (c) Apply the inverse of the marginal utility function to the interpolated expected marginal value to find the policy function of consumption $\hat{c}_{it}$ on the endogenous grid $\tilde{k}_{it}$.
 - (d) Compute the endogenous grid points $\tilde{k}_{it}$ from the budget constraint given the policy $\hat{c}_{it}$.
 - (e) Interpolate the consumption policy function $\hat{c}_{it}$ from the endogenous grid $\tilde{k}_{it}$ onto the exogenous grid $\vec{k}$ to obtain an updated policy function c_{it}^m .
 - (f) Enforce the borrowing constraint.
 - (g) Check convergence by verifying for a small ϵ , whether the condition $|u'^{-1} \left(\frac{\partial V_{it}^m}{\partial k_{it}} \right) - u'^{-1} \left(\frac{\partial V_{it}^{m-1}}{\partial k_{it}} \right)| < \epsilon$ is true. $|\cdot|$ denotes the Euclidean norm. If the condition is not satisfied, repeat steps (b)–(g).
4. With the converged global policy functions, we aggregate and simulate the economy using stochastic simulation of [Young \(2010\)](#) for T periods:
- (a) We set the initial capital stock to the value from the deterministic steady state of a representative agent economy and denote the capital stock as K_1 .¹⁵
 - (b) In period t , we have capital stock K_t , which is generally off-grid. To evaluate the policy functions of the household, we evaluate the individual policy function at K_t by interpolating from the exogenous grid $\vec{K}$ on the current capital stock K_t . Evaluate the policy functions at the current aggregate state $\{A_t, \zeta_t, \delta_t\}$.
 - (c) Given household policies evaluated at the state realizations today, we update the household distribution using the stochastic simulation method of [Young \(2010\)](#).
 - (d) Repeat steps (b) and (c) for T periods.

¹⁴ Note that the idiosyncratic risk depends on the realization of the aggregate risk.

¹⁵ We also find a distribution that has the mean of K_1 and initialize the simulation with this distribution.

5. Discard the first 1000 periods as a burn-in sample. Use the remaining time series to update the perceived law of motion by regressing the logarithm of the capital stock on the aggregate states and the lagged logarithm of the capital stock as in equation (92). Denote the resulting regression coefficients as $\tilde{\beta}^m$
6. Check whether $|\tilde{\beta}^n - \beta^{n-1}| < \epsilon$ for a small ϵ . If the condition is met, stop; otherwise, update the coefficients as $\beta^n = \varphi \tilde{\beta}^n + (1 - \varphi) \beta^{n-1}$ with $\varphi \in (0, 0.5)$ and repeat steps (3) - (6).

Note that the description above accounts for all aggregate shocks but also accommodates cases with fewer aggregate shocks by keeping some of them fixed. We apply steps (1) to (6) to each satellite model, generating a dataset that is then used to train the surrogate model.